Supplement to "A fast elementary series to compute power-law limb-darkened transiting exoplanet light curves"

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This supplement contains material that is not needed to use or to check the default method of the paper: the error bounds for the truncation of the interior series by powers of p (Section S1), the construction, validation and benchmarks of the contact expansions about first and second contact and of their certified pieces, which Sect. 6.3 of the paper summarises (Sections S2 and S3), and the proof of the bound for the contact pieces, Theorem 1 (Section S4). Equation numbers with a prefix S refer to this supplement. Readable explanations of all proofs, and notebooks that check every bound numerically against references independent of the derivations, are in the same repository (`docs/proofs` and `validation`).

Notation from the paper

The stellar disk is the unit disk, r the distance from its centre, $\mu = \sqrt{1-r^2}$, and $D_\gamma(z,p) = \langle (1-r^2)^\gamma \rangle$ the mean of $(1-r^2)^\gamma$ over an occulting disk of radius p centred at distance z; $w = 1-z^2$, $v = 1/w$, $\beta_m = (-1)^m \binom{\gamma}{m}$. Truncated by powers of p, the series of the paper is $S_N^{\mathrm{P}} = \sum_{n \le N} T_n$, where T_n is the coefficient of p^{2n}, equal to the n-th term of the mean-value expansion

$$\langle f \rangle = \sum_{n=0}^{\infty} \frac{(p^2/4)^n}{n!\,(n+1)!}\, \Delta^n f\big|_{\mathrm{centre}}, \tag{S1}$$

with Δ the Laplacian in the sky plane. Across the limb and inside the disk, the blocked integral $P_\gamma = \int (1-r^2)^\gamma \,\mathrm{d}A$ over the occulted part of the star is, with $s = r^2$, $a = (z-p)^2$, $b = (z+p)^2$, $s_1 = 1$ across the limb and $s_1 = b$ inside,

$$P_\gamma = -\frac{1}{2(\gamma+1)} \int_a^{s_1} (1-s)^{\gamma+1} \left[1 + \frac{p^2-z^2}{s}\right] \frac{\mathrm{d}s}{\sqrt{(s-a)(b-s)}}, \tag{S2}$$

With $x = 1-(z-p)^2$, $c_0 = 1+p^2-z^2$, $c_j = (p^2-z^2)x^j$ $(j \ge 1)$,

$$k = \frac{1-(z-p)^2}{4zp}, \tag{S3}$$

and $\kappa = 1/k$,

$$P_\gamma = -\frac{x^{\gamma+1}\sqrt{k}}{2(\gamma+1)} \sum_{j \ge 0} c_j\, B\left(\tfrac{1}{2}, \gamma+2+j\right)\, {}_2F_1\left(\tfrac{1}{2}, \tfrac{1}{2}; \gamma+\tfrac{5}{2}+j; k\right), \tag{S4}$$

$$P_\gamma = -\frac{\pi\, x^{\gamma+1}}{2(\gamma+1)} \sum_{j \ge 0} c_j\, {}_2F_1\left(-\gamma-1-j, \tfrac{1}{2}; 1; \kappa\right). \tag{S5}$$

For fixed p, z is an algebraic function of k,

$$z = p(1-2k) + \sqrt{p^2(1-2k)^2 + 1 - p^2}, \tag{S6}$$

Theorem 1 (Contact pieces) *Let $p < \frac{1}{2}$. On every piece of the contact expansions, $|P_\gamma - \hat{P}_\gamma| \le E_\gamma^{(j)} + E_\gamma^{(n)}$, where $\hat{P}_\gamma$ is the polynomial approximation of the piece, $E_\gamma^{(j)}$ (Equation S24) bounds the remainder of the j-series and $E_\gamma^{(n)}$ (Equations S29–S30) the truncation in n. Both are explicit functions of p, γ, J, N, the piece and a radius ρ, and are finite whenever the disk conditions of Lemma 2 hold. The flux error is at most $\sum_i |c_i|\,(E_{\gamma_i}^{(j)} + E_{\gamma_i}^{(n)})/\big(2\pi \sum_i c_i/(\alpha_i+2)\big)$.*

S1 Truncation by powers of p

Truncation by powers of p gives a second family of approximations, $S_N^{\mathrm{P}} = \sum_{n \le N} T_n$, which is also a polynomial in v, of degree $2N$. Two bounds apply to it.

Proposition 1 (Mean-value remainder) *Let $f \in C^{2N+2}$ on the closed occulting disk $\bar{D}$. Then*

$$\langle f \rangle - \sum_{n=0}^{N} \frac{(p^2/4)^n}{n!(n+1)!} \Delta^n f\big|_{\mathrm{centre}} = \frac{(p^2/4)^{N+1}}{(N+1)!\,(N+2)!}\, \Delta^{N+1} f(\xi) \tag{S7}$$

for some $\xi \in \bar{D}$. For $f = (1-r^2)^\gamma$ with $0 < \gamma < 1$, $\Delta^n f \le 0$ and $|\Delta^n f|$ increases with r for $n \ge 1$. The remainder therefore lies between the right-hand side evaluated at $r = \max(0, z-p)$ and at $r = z+p$.

The proof, which follows Pizzetti (1909), is given in Section S1.1. The upper end of this two-sided bound grows like $(p/(1-z-p))^{2N+2}$ and is useful only for $z < 1-2p$. A bound valid up to contact follows from the sign structure.

Proposition 2 (Abel bound) *Let $\gamma = \frac{1}{2}$, $t = p^2$ and $t_1 = (1-z)^2$. Then for $0 \le p \le 1-z$,*

$$0 \le S_N^{\mathrm{P}}(z,p) - D_{1/2}(z,p) \le \left(\frac{p}{1-z}\right)^{2N+2} \left[S_N^{\mathrm{P}}(z,1-z) - D_{1/2}(z,1-z)\right], \tag{S8}$$

where the internally tangent disk has the elementary mean

$$D_{1/2}(1-q,q) = \frac{2\arccos(1-2q) - \frac{4}{3}(3+2q-8q^2)\sqrt{q(1-q)}}{3\pi q^2}, \qquad 0 < q < 1. \tag{S9}$$

Proof. As a power series in t, $D_{1/2}(z,\sqrt{t}) = \sum_n d_n t^n$ with $d_n \le 0$ for $n \ge 1$, convergent for $t < t_1$. The partial sums at t_1 are non-increasing in N and bounded below by $D_{1/2}(z,1-z)$, because $S_N^{\mathrm{P}}(z,\sqrt{t}) \ge D_{1/2}(z,\sqrt{t})$ for $t < t_1$ and $D_{1/2}$ is continuous in p. The series therefore converges at t_1, and by Abel's theorem its sum is $D_{1/2}(z,1-z)$. Then $\sum_{n>N} |d_n| t^n \le (t/t_1)^{N+1} \sum_{n>N} |d_n| t_1^n$. Equation (S9) is the $b + r = 1$ case of Agol et al. (2020), divided by q^2; it agrees with direct quadrature to 10^{-27}. □

S1.1 Mean-value expansion with remainder on the disk

Let $f \in C^{2N+2}(\bar{D})$, where $\bar{D}$ is the closed disk of radius p about c, and let $m_g(\rho)$ denote the mean of g over the circle of radius ρ about c. Green's theorem gives

$$2\pi\rho\, m_f'(\rho) = \int_{|x-c|\le\rho} \Delta f \,\mathrm{d}A = 2\pi \int_0^\rho s\, m_{\Delta f}(s)\,\mathrm{d}s. \tag{S10}$$

Integrating from 0 gives

$$m_f(\rho) = f(c) + (\mathcal{T} m_{\Delta f})(\rho), \qquad (\mathcal{T} g)(\rho) = \int_0^\rho s \ln\frac{\rho}{s}\, g(s)\,\mathrm{d}s. \tag{S11}$$

The operator $\mathcal{T}$ has a non-negative kernel and maps ρ^{2j} to $\rho^{2j+2}/(2j+2)^2$. Iterating, with $m_{\Delta^j f}(0) = \Delta^j f(c)$,

$$m_f(\rho) = \sum_{j=0}^{N} \frac{\rho^{2j}}{4^j (j!)^2} \Delta^j f(c) + (\mathcal{T}^{N+1} m_{\Delta^{N+1} f})(\rho). \tag{S12}$$

The area mean is $\langle f\rangle = (2/p^2)\int_0^p \rho\, m_f(\rho)\,\mathrm{d}\rho$. The polynomial terms give the coefficients $(p^2/4)^j/(j!(j+1)!)$ of Equation (S1). The remainder is $\int_0^p K(s)\, m_{\Delta^{N+1} f}(s)\,\mathrm{d}s$ with a non-negative kernel K, whose integral equals the remainder for $\Delta^{N+1} f \equiv 1$, namely $(p^2/4)^{N+1}/((N+1)!(N+2)!)$. Since each circle mean lies between the minimum and maximum of $\Delta^{N+1} f$ on $\bar{D}$, the mean-value theorem for integrals gives Equation (S7).

For $f = (1-r^2)^\gamma = \sum_m \beta_m r^{2m}$ with $0 < \gamma < 1$, $\Delta^n r^{2m} = 4^n\,(m!/(m-n)!)^2\, r^{2(m-n)}$ for $m \ge n$. Hence $\Delta^n f = \sum_{m\ge n} \beta_m 4^n (m!/(m-n)!)^2 r^{2(m-n)}$ has only negative coefficients for $n \ge 1$. It is therefore negative, and its modulus increases with r. On the disk r ranges over $[\max(0, z-p), z+p]$ (for $z < p$ the disk covers the stellar centre), which gives the two-sided bound of Proposition 1.

S2 Expansions about the contacts

This section and the next give the construction of the contact expansions and of their certified pieces, which Sect. 6.3 of the paper summarises. We keep the notation of the paper: the exponent is $\alpha = 2\gamma$, ζ_j denotes the coefficients c_j of Equations (S4) and (S5), $y = 1-k$ and $y' = 1-\kappa$ measure the distance from second contact outside and inside the stellar disk, and

$$\begin{aligned} \mathcal{H}_j(k) &= B(\tfrac{1}{2}, \tfrac{\alpha}{2}+2+j)\, {}_2F_1(\tfrac{1}{2}, \tfrac{1}{2}; \tfrac{\alpha+5}{2}+j; k) \\ \mathcal{G}_j(\kappa) &= \pi\, {}_2F_1(-\tfrac{\alpha}{2}-1-j, \tfrac{1}{2}; 1; \kappa), \end{aligned} \tag{S13}$$

which depend solely on α, j, and their argument, so that $P_\alpha = -x^{\alpha/2+1}\sqrt{k}\,(\alpha+2)^{-1}\sum_j \zeta_j \mathcal{H}_j(k)$ across the limb and $P_\alpha = -x^{\alpha/2+1}(\alpha+2)^{-1}\sum_j \zeta_j \mathcal{G}_j(\kappa)$ inside the disk; in the notation of Sect. S4, $\mathcal{H}_j = H_j$ and $\mathcal{G}_j = \pi G_j$ (Equation S23). Both $\mathcal{H}_j(k)$ and $\mathcal{G}_j(\kappa)$ are positive and monotonically non-increasing with j, while every coefficient ζ_j with $j \ge 1$ shares the sign of $p^2 - z^2$. Consequently, the truncation remainder of the j-series after any index $J \ge 1$ is bounded in magnitude by the first omitted term divided by $1-x$ (Lemma 1).

Second contact corresponds to a unit argument ($k = 1$ or $\kappa = 1$) of these Gauss functions. A Gauss function ${}_2F_1(\hat{a}, \hat{b}; \hat{c}; 1-y)$ with parameter excess $\nu = \hat{c} - \hat{a} - \hat{b} > 0$ is finite but non-analytic as $y \to 0^+$; for non-integer ν, the connection formula (Olver et al., 2010, 15.8.4) gives

$$\begin{aligned} {}_2F_1(\hat{a}, \hat{b}; \hat{c}; 1-y) = y^\nu \frac{\Gamma(\hat{c})\Gamma(-\nu)}{\Gamma(\hat{a})\Gamma(\hat{b})}\, {}_2F_1(\hat{c}-\hat{a}, \hat{c}-\hat{b}; 1+\nu; y) \\ + \frac{\Gamma(\hat{c})\Gamma(\nu)}{\Gamma(\hat{c}-\hat{a})\Gamma(\hat{c}-\hat{b})}\, {}_2F_1(\hat{a}, \hat{b}; 1-\nu; y). \end{aligned} \tag{S14}$$

For integer ν, the two series merge into a finite polynomial in y plus $y^\nu \ln y$ times a power series (Olver et al., 2010, 15.8.10). For both $\mathcal{H}_j$ and $\mathcal{G}_j$, the parameter excess is $\nu = \frac{\alpha+3}{2} + j$, an integer if and only if α is an odd integer. For any $\alpha > 0$, we may therefore write $\mathcal{H}_j$ in the unified form

$$\mathcal{H}_j = U_j(y) + y^{(\alpha+3)/2+j}\big[L_j(y)\ln y + C_j(y)\big], \qquad y = 1-k, \tag{S15}$$

where U_j, L_j, and C_j are convergent power series in y. Note that Eq. (S15) holds for all $\alpha > 0$, with the logarithmic term appearing only when α is an odd integer: specifically, if α is not an odd integer, then ν is non-integer, $L_j(y) \equiv 0$, and $U_j(y)$ and $C_j(y)$ correspond directly to the two hypergeometric series in Eq. (S14); whereas if α is an odd integer, $U_j(y)$, $L_j(y)$, and $C_j(y)$ are given by the logarithmic connection formula (Olver et al., 2010, 15.8.10). The interior function $\mathcal{G}_j(\kappa)$ admits an identical decomposition in terms of $y' = 1-\kappa$.

For a fixed radius ratio p, the separation z is the algebraic function of k given by Equation (S6). Consequently, $x = 4zpk$, the coefficients ζ_j, and the prefactors in Eqs. (S4) and (S5) are all analytic functions of k in neighbourhoods of both $k = 0$ and $k = 1$. Expanding each factor in power series alongside the Gauss hypergeometric functions and collecting like powers yields three contact expansions covering the outer and inner neighbourhoods of the stellar limb:

$$P_\alpha = k^{(\alpha+3)/2}\sum_{n\geq 0} e_n k^n, \tag{S16}$$

$$P_\alpha = \sum_{n\geq 0} g_n y^n + y^{(\alpha+3)/2}\sum_{n\geq 0}\left(g'_n \ln y + g''_n\right) y^n, \tag{S17}$$

$$P_\alpha = \sum_{n\geq 0} h_n y'^n + y'^{(\alpha+3)/2}\sum_{n\geq 0}\left(h'_n \ln y' + h''_n\right) y'^n, \tag{S18}$$

valid for $0 \leq k \leq \frac{1}{2}$, $0 \leq y \leq \frac{1}{2}$, and $0 \leq y' \leq \frac{1}{2}$, respectively. Here $y = 1-k$ and $y' = 1-\kappa$ measure the dimensionless distance from second contact on the exterior ($z \geq 1-p$) and interior ($z \leq 1-p$) sides of the stellar limb. Just like Eq. (S15), Eqs. (S17) and (S18) apply to every $\alpha > 0$: when α is not an odd integer, $g'_n = h'_n = 0$ and the logarithmic terms vanish; when α is an odd integer, $y^{(\alpha+3)/2}$ reduces to an integer power of y, and the logarithmic series carries the non-analytic part of the flux.

To compute the expansion coefficients, we express every factor in Eqs. (S4) and (S5) as a power series in the local expansion variable $\eta \in \{k, y, y'\}$ and multiply the resulting series. Substituting $k(\eta)$, with $k = 1-y$ or $k = 1/(1-y')$, into Eq. (S6) defines $z(\eta)$, which in turn determines $x(\eta) = 1-(z(\eta)-p)^2$, the coefficients $\zeta_j(\eta)$, and the overall prefactor

$$\Phi(\eta) = -\frac{x(\eta)^{\alpha/2+1}\sqrt{k(\eta)}}{\alpha+2}\ \text{(limb)}, \quad \Phi(\eta) = -\frac{x(\eta)^{\alpha/2+1}}{\alpha+2}\ \text{(inside)}, \tag{S19}$$

such that $P_\alpha = \Phi(\eta)\sum_j \zeta_j \mathcal{H}_j$ across the limb and $P_\alpha = \Phi(\eta)\sum_j \zeta_j \mathcal{G}_j$ inside the disk. For fixed p and α, all of these quantities are functions of the single variable η. The coefficients in Eqs. (S16)–(S18) are then defined as follows:

1. e_n: the Taylor coefficients in k of $-\frac{(4pz(k))^{\alpha/2+1}}{\alpha+2}\sum_j \zeta_j(k)\,\mathcal{H}_j(k)$, where the leading fractional power $k^{(\alpha+3)/2}$ has been factored out using the identity $x^{\alpha/2+1}\sqrt{k} = (4pz)^{\alpha/2+1}k^{(\alpha+3)/2}$ from Equation (S3).

2. g_n, g'_n, and g''_n: the Taylor coefficients in y of $\Phi(y)\sum_j \zeta_j(y)U_j(y)$, $\Phi(y)\sum_j \zeta_j(y)\,y^j L_j(y)$, and $\Phi(y)\sum_j \zeta_j(y)\,y^j C_j(y)$, respectively, where U_j, L_j, and C_j are the component series in

Eq. (S15), and $\Phi(y)$ and $\zeta_j(y)$ are evaluated via $z(k = 1 - y)$ in Eq. (S6); here the factor $y^{(\alpha+3)/2+j}$ in Eq. (S15) has been split into the common prefactor $y^{(\alpha+3)/2}$ of Eq. (S17) and the factor y^j inside the sum.

3. h_n, h'_n, and h''_n: defined identically to g_n, g'_n, and g''_n, upon replacing y with y', $\mathcal{H}_j$ with $\mathcal{G}_j$, and the limb prefactor with the interior prefactor in Eq. (S19).

At first contact ($k = 0$), for instance, one has $z = 1 + p$, $x = 0$, $\zeta_0 = 1 + p^2 - z^2 = -2p$, and $\zeta_j = 0$ for all $j \geq 1$, yielding the closed-form leading coefficient

$$\begin{aligned} e_0 &= \frac{2p\,\big(4p(1+p)\big)^{\alpha/2+1}}{\alpha+2}\,B\big(\tfrac{1}{2},\tfrac{\alpha}{2}+2\big)\,, \\ P_\alpha &\simeq e_0\,k^{(\alpha+3)/2} \qquad (k \to 0). \end{aligned} \tag{S20}$$

Higher-order coefficients depend solely on p and α, and — truncating all power series at order N in η and the hypergeometric sum at $j \leq J$ (Sect. S3) — are constructed in four steps of elementary power-series arithmetic:

(i) Expand $z(\eta)$ from Eq. (S6) by taking the square root of a power series, and form $x(\eta)$ and $\zeta_j(\eta)$ by series multiplication.

(ii) Construct the prefactor $\Phi(\eta)$ by raising power series to real powers: given a series $\psi(\eta) = \sum_{i\geq 0}\psi_i\eta^i$ with leading coefficient $\psi_0 > 0$, the coefficients of $\psi(\eta)^\lambda = \sum_{n\geq 0}\psi_n^{(\lambda)}\eta^n$, for any real exponent λ, satisfy Euler's recurrence $\psi_0^{(\lambda)} = \psi_0^\lambda$ and $\psi_n^{(\lambda)} = \frac{1}{n\psi_0}\sum_{i=1}^{n}\big[(\lambda+1)i-n\big]\psi_i\psi_{n-i}^{(\lambda)}$ for $n \geq 1$ (Knuth, 1997, Sect. 4.7).

(iii) Assemble the power-series coefficients of the Gauss hypergeometric functions, which are available in closed form. Around $k = 0$, $\mathcal{H}_j(k) = \sum_n \mathcal{H}_{j,n}k^n$ with $\mathcal{H}_{j,n} = B(\frac{1}{2},\frac{\alpha}{2}+2+j)\,(\frac{1}{2})_n^2/[(\frac{\alpha+5}{2}+j)_n\,n!]$, where $(\cdot)_n$ denotes the rising factorial (Pochhammer symbol). Around second contact, for non-integer $\nu = \frac{\alpha+3}{2}+j$, applying Eq. (S14) with $(\hat{a},\hat{b},\hat{c}) = (\frac{1}{2},\frac{1}{2},\frac{\alpha+5}{2}+j)$ expresses $U_j(y)$ and $C_j(y)$ as $B(\frac{1}{2},\frac{\alpha}{2}+2+j)$ times the Gamma-function prefactors of Eq. (S14) multiplied by power series with coefficients $(\frac{1}{2})_n^2/[(1-\nu)_n\,n!]$ and $(\frac{\alpha}{2}+2+j)_n^2/[(1+\nu)_n\,n!]$, respectively. The interior functions $\mathcal{G}_j$ follow analogously with $(\hat{a},\hat{b},\hat{c}) = (-\frac{\alpha}{2}-1-j,\frac{1}{2},1)$, while the logarithmic case for odd integer α follows directly from Olver et al. (2010, 15.8.10).

(iv) Multiply the component series from steps (i)–(iii) and accumulate the sum over $j \leq J$.

Geometrically, Eq. (S16) covers the outer half of ingress/egress near first/fourth contacts ($\sqrt{1-p^2} \leq z \leq 1+p$), Eq. (S17) covers the inner half between second/third contact and $z = \sqrt{1-p^2}$, and Eq. (S18) covers the region immediately inside second/third contacts ($\sqrt{1+8p^2}-3p \leq z \leq 1-p$). In particular, Eq. (S16) demonstrates that the flux deficit vanishes at first/fourth contact as $(1+p-z)^{(\alpha+3)/2}$, while Eqs. (S17) and (S18) explicitly isolate the non-analytic behaviour at second/third contacts, where both y and y' scale linearly with $|1-p-z|$.

Once these singular factors are extracted explicitly, the remaining power series in k, y, and y' converge geometrically. Because the nearest singularity in the complex plane lies either at the opposite contact ($k = 1$ for Eq. S16 and $y = 1$ for Eq. S17) or where the occulter rim crosses the stellar centre ($y' = 1 - 4p^2$ for Eq. S18), consecutive terms decay by a factor of at least $\approx \frac{1}{2}$ across the stated intervals. Truncating at degree $n = 32$ therefore achieves 10^{-15} accuracy (Table S1), with the n-truncation residual estimated in fixed-order mode from the magnitude of the last three terms and bounded formally in Sect. S3. In contrast, the auxiliary sum over j converges more slowly as

$x \to 1$ ($z \to p$). Retaining at most $J = 200$ terms in j yields a relative accuracy of 10^{-11} for all $p \leq 0.2$; for larger occulters the expansions do not reach the tolerance, and the fixed-node rule of Sect. 6.2 of the paper ($p < 0.3$) or numerical quadrature takes over. Importantly, for $M = 11$ and every configuration listed in Table C.3 of the paper, the interior cut-off z_c lies well above the lower boundary $\sqrt{1 + 8p^2} - 3p$ of Eq. (S18); consequently, the interior polynomial and the three contact expansions together provide seamless coverage across all phases of any transit with $p \leq 0.2$.

Note that the integer exponents $\alpha_i = 0$ and $\alpha_i = 2$ require no hypergeometric expansion, as their blocked flux integrals reduce to the elementary circle–circle intersection area and an algebraic function of z and p, respectively. For each remaining exponent, precomputing the expansion coefficients requires $O(JN^2)$ operations per light curve for J terms in j and degree $N = 32$, amounting to 0.1–1.7 ms per light curve for the standard laws in Table S1. Evaluating each limb sample then requires only one logarithm, one fractional power, and one to three degree-32 Horner polynomial evaluations per exponent.

We validated the contact expansions of Eqs. (S16)–(S18) against 40-digit evaluations of Equation (S2) and of the area integral of μ^α over the occulter across exponents $\alpha \in \{0, \frac{1}{2}, 0.6, 1, \frac{3}{2}, 2, \frac{5}{2}\}$ and radius ratios $0.001 \leq p \leq 0.2$, extending to separations within 10^{-12} of first and second contact. Across all tests, the maximum error in normalised flux units was 1.1×10^{-15}. When evaluated through the full light-curve pipeline on $z_c < z < 1+p$, the square-root, four-parameter, and power-2 laws agree with an independent 40-digit Green's-theorem quadrature to better than 10^{-15} in fixed-order mode, and remain strictly within τ in the certified piecewise mode of Sect. S3 (Table S1). After a one-time set-up of 0.1–1.7 ms per light curve, the fixed-order contact expansions evaluate each limb sample 15 to 22 times faster than Green's-theorem quadrature—and 50 to 300 times faster than summing the single-index expansion of Eq. (S4) directly at each point (Sect. 4 of the paper). This acceleration arises because the contact expansions rearrange the double sum once per light curve, folding both the sum over j and the hypergeometric power series into the precomputed coefficients $(e_n, g_n, g'_n, g''_n, h_n, h'_n, h''_n)$ so that no special function is evaluated at sample time.

S3 Certified pieces

To obtain a guaranteed upper bound on the truncation error of the contact expansions in Eqs. (S16)–(S18), we partition the limb region $z_c < z < 1+p$ into a small number of subdomains ("pieces"), on each of which a low-degree polynomial suffices, and bound on every piece both the tail of the j-series beyond $j = J$ and the tail of the power series in n beyond $n = N$. The construction is given below; the resulting error bound is Theorem 1, proven in Sect. S4. For a target tolerance of $\tau = 10^{-9}$, every piece is certified for all $p \leq 0.2$ using five to seven pieces of polynomial degree $N \in [2, 13]$; for $\tau = 10^{-12}$, certification succeeds for $p \leq 0.15$ using five to eight pieces of degree up to $N = 17$. Any pieces that cannot be certified—such as those for larger occulters, or the innermost pieces at the tightest tolerance where the j-series converges slowly as $z \to p$—automatically fall back to numerical quadrature. Consequently, for all $p \leq 0.2$ and $\tau = 10^{-9}$, the interior series combined with the certified contact pieces evaluates the entire transit light curve with a mathematically proven error bound of τ. In validation tests against 40-digit reference integrations, none of the 355 certified pieces violated its theoretical bound, with actual maximum errors ranging between 0.56 and 0.98 times the bound (see below).

The computational overhead of certifying the piecewise error bound is incurred only once per light curve. At $\tau = 10^{-9}$, this set-up requires 5–64 ms (Table S1), compared with 0.1–1.7 ms for the uncertified fixed-order coefficients; the majority of the certification time is spent computing the $J \leq 200$ terms needed for the intermediate Taylor pieces and evaluating the complex-circle

Table S1: Performance and accuracy of the contact expansions (Sect. S2) across $z_c < z < 1 + p$, paired with the interior series run at the same tolerance τ: maximum absolute flux error against a 40-digit reference across 34–40 separations (including points within $10^{-13}p$ of the contacts), one-time coefficient set-up time per light curve (summing all exponents of the law), and evaluation time per limb sample. "Fixed order": Equations (S16)–(S18) truncated at $n = 32$ with empirical tail estimation (Sect. S2). "Proven": the certified piecewise polynomials of Sect. S3, each guaranteed to satisfy $|\Delta F| \leq \tau$. "Quadr.": Green's-theorem numerical quadrature. Benchmarks run on a single thread of an Intel Xeon (2.1 GHz) with `numba` 0.68.

		fixed order			proven, $\tau = 10^{-9}$			proven, $\tau = 10^{-12}$			quadr.
Law	p	max $\lvert\Delta F\rvert$	set-up (ms)	per sample (ns)	max $\lvert\Delta F\rvert$	set-up (ms)	per sample (ns)	max $\lvert\Delta F\rvert$	set-up (ms)	per sample (ns)	per sample (ns)
square-root	0.0092	$< 10^{-16}$	0.17	77	3.2×10^{-10}	9	54	2.8×10^{-13}	13	1419	
square-root	0.1	1.1×10^{-16}	0.44	80	2.7×10^{-10}	19	66	1.6×10^{-13}	41	1277	
square-root	0.2	5.6×10^{-16}	0.91	78	1.6×10^{-10}	27	54	1.4×10^{-13} [a]	34	1143	
four-parameter	0.0092	$< 10^{-16}$	0.26	77	5.3×10^{-10}	16	87	2.8×10^{-13}	21	1648	
four-parameter	0.1	1.1×10^{-16}	0.63	81	1.1×10^{-10}	23	72	6.7×10^{-13}	75	1403	
four-parameter	0.2	1.1×10^{-16}	1.68	80	5.7×10^{-11}	64	75	2.2×10^{-13} [a]	61	1758	
power-2, $\alpha = 0.6$	0.0092	$< 10^{-16}$	0.09	54	4.0×10^{-10}	5	40	3.8×10^{-13}	7	1132	
power-2, $\alpha = 0.6$	0.1	1.1×10^{-16}	0.17	61	4.0×10^{-10}	12	51	1.8×10^{-13}	22	1307	
power-2, $\alpha = 0.6$	0.2	2.2×10^{-16}	0.40	62	7.1×10^{-10}	17	52	1.3×10^{-13} [a]	25	1017	
power-2, $\alpha = 1$	0.0092	$< 10^{-16}$	0.08	60	2.7×10^{-10}	5	47	3.0×10^{-13}	7	1083	
power-2, $\alpha = 1$	0.1	1.1×10^{-16}	0.20	49	2.7×10^{-10}	12	43	1.5×10^{-13}	26	977	
power-2, $\alpha = 1$	0.2	7.8×10^{-16}	0.52	58	4.1×10^{-10}	22	51	1.5×10^{-13} [a]	29	1000	

[a] 76–77 % of the limb samples certified; the two innermost pieces ($z_c < z < 0.625$) cannot reach 10^{-12} within the cap of $J = 200$ terms of the j-series and automatically fall back to quadrature.

majorants $\mathcal{M}(R)$. Once constructed, evaluating each sample on the certified pieces takes 40–87 ns—comparable to the 49–81 ns of the fixed-order expansions, and 19 to 29 times faster than numerical quadrature. For a complete transit light curve of 10^4 samples at $p = 0.1$, the one-time set-up dominates the total runtime in certified mode: 12 ms in total for the power-2 law, compared with 0.5 ms using the fixed-order expansions and 3.2 ms using `batman`. Accordingly, our software implementation employs uncertified limb methods by default (the fixed-node rule of Sect. 6.2 of the paper and its Chebyshev interpolants) and enables the certified piecewise bounds on user request.

The total truncation error of the contact expansions, Eqs. (S16)–(S18), comprises two contributions: the tail of the j-series beyond $j = J$, which is bounded geometrically as noted in Sect. S2 (Lemma 1), and the tail of the power series in n beyond $n = N$. To bound the latter provably while keeping the polynomial degree small, we partition the limb region $z_c < z < 1 + p$ into subdomains ("pieces") on each of which a low-degree polynomial suffices: specifically, Eq. (S16) on $k \leq \frac{1}{4}$; two analytic Taylor expansions centred at $k = \frac{3}{8}$ and $k = \frac{5}{8}$; Eq. (S17) on $y \leq \frac{1}{4}$; Eq. (S18) on $y' \leq \frac{1}{4}$ (narrowed if z_c lies closer); and analytic Taylor expansions in κ of half-width at most 0.07 extending inward from there to $\kappa_c = \kappa(z_c)$. On any piece with local coordinate η and half-width (or contact range) ω, the remainder of each analytic series $\sum_n f_n \eta^n$ satisfies

$$\Big|\sum_{n>N} f_n \eta^n\Big| \leq \sum_{n=N+1}^{64} |f_n|\, \omega^n + \mathcal{M}(R)\, \frac{(\omega/R)^{65}}{1 - \omega/R}, \qquad |\eta| \leq \omega < R, \tag{S21}$$

where the coefficients f_n are computed explicitly through $n = 64$ and higher terms ($n \geq 65$) are bounded via Cauchy's inequality $|f_n| \leq \mathcal{M}(R) R^{-n}$, with $\mathcal{M}(R)$ an analytical upper bound on $|f(\eta)|$ along the circle $|\eta| = R$ in the complex plane. On the contact pieces, the analytic series are the component power series in Eqs. (S16)–(S18), whose tails are weighted by at most $\omega^{(\alpha+3)/2}$ and $\max_{0<y\leq\omega} y^{(\alpha+3)/2} |\ln y|$ across the piece. To select the truncation degree N on each piece, we weight

both the j-truncation and n-truncation bounds by $|c_i|$, sum over all exponents of the limb-darkening law, and choose the smallest N for which the total guaranteed flux error does not exceed the target tolerance τ. The resulting bound is stated in Theorem 1.

The complete proof of Theorem 1, together with explicit formulae for the majorants $\mathcal{M}(R)$ and the complex-disk analyticity conditions on $|\eta| \le R$, is given in Sect. S4. The proof relies on complex and real analysis: the j-remainder bound follows from the monotonicity of the integrand in Eq. (S2); $z(\eta)$, $x(\eta)$, and the prefactors $\Phi(\eta)$ are holomorphic on the disks $|\eta| \le R$ because z is non-vanishing and all square-root arguments maintain a positive real part; their maxima on $|\eta| = R$ are bounded rigorously from 1024 boundary samples combined with a Cauchy derivative bound; and the Gauss hypergeometric series are bounded via majorant series with geometric tails, Euler's integral representation, and — for odd integer α — elementary inequalities for the digamma function. On the intermediate Taylor pieces, the expansion coefficients are formed by re-centring a degree-64 series about the piece midpoint, and the error bound explicitly accounts for that finite re-centring truncation. While Theorem 1 holds in exact arithmetic, our implementation evaluates the bound in IEEE-754 double precision and inflates it by a safety factor of $1 + 10^{-6}$; floating-point round-off in the coefficients and Horner evaluation contributes only $\sim 10^{-16}$. The Cauchy estimate in Eq. (S21) is valid only if the expanded function is analytic on the whole complex disk $|\eta| \le R$ around the piece, which requires that $z(\eta)$ does not vanish and that every square-root argument keeps a positive real part on that disk. The implementation verifies these conditions numerically for each piece; if they cannot be verified for any admissible radius r, no bound is claimed for that piece, and its samples are computed by quadrature instead.

As z approaches p, $x \to 1$ and the j-series converges only as x^J. With our cap of $J = 200$ terms, the innermost pieces therefore cannot reach $\tau = 10^{-12}$ at $p = 0.2$, and certain pieces fall short of $\tau = 10^{-9}$ for $p \ge 0.25$; whenever a piece is not certified, the implementation falls back to numerical quadrature for the affected samples. To assess the sharpness of Theorem 1, we tested each certified piece individually against 40-digit reference integrations across 29 separations spanning the piece (including both endpoints). Across 60 test configurations — covering the square-root, four-parameter, and three power-2 laws ($\alpha = 0.6$, 1, and 1.7), radius ratios p from 0.0092 to 0.3, and tolerances $\tau = 10^{-9}$ and 10^{-12} — none of the 355 certified pieces violated its theoretical bound, and the maximum ratio of actual error to proven bound in each configuration ranged between 0.56 and 0.98. The bound is therefore remarkably tight, with the certified pieces exhibiting actual errors of $\approx \tau/2$, compared with $\sim 10^{-15}$ for the degree-32 fixed-order expansions (Table S1).

S4 Proof of the bound for the contact pieces

This section proves Theorem 1 (stated in the notation section above). Throughout, $0 < p < \frac{1}{2}$, $m = \gamma + \frac{3}{2}$, and u is the variable of a piece: $u = k$ (first contact), $u = y = 1 - k$ and $u = y' = 1 - \kappa$ (second contact), $u = k - k_0$ and $u = \kappa - \kappa_0$ (middle pieces centred at k_0 and κ_0). On each piece the variable satisfies $|u| \le w$. For a power series $F(u) = \sum_n t_n u^n$ we write $\hat{F}(r) = \sum_n |t_n| r^n$ and $\hat{F}^{>n_0}(r) = \sum_{n>n_0} |t_n| r^n$.

S4.1 Representation

In Equation (S2) substitute $s = 1 - x(1 - t)$. Then $s - a = xt$, $b - s = 4zp(1 - kt)$ because $b - 1 + x = b - a = 4zp$, and $\mathrm{d}s = x\,\mathrm{d}t$, so that

$$P_\gamma = -\frac{x^{\gamma+1}\sqrt{k}}{2(\gamma+1)} \int_0^{t_1} t^{-1/2}(1-t)^{\gamma+1}(1-kt)^{-1/2} \left[1 + \frac{p^2 - z^2}{s}\right] \mathrm{d}t, \tag{S22}$$

with $t_1 = 1$ across the limb and $t_1 = 4zp/x = \kappa$ inside the disk. For $z > p$ we have $0 < x < 1$ and $0 \le x(1-t) \le x$, so $1/s = \sum_j x^j (1-t)^j$ converges uniformly on $[0,1]$ and may be integrated term by term. With

$$H_j(k) = \int_0^1 t^{-1/2}(1-t)^{\gamma+1+j}(1-kt)^{-1/2}\mathrm{d}t, \qquad G_j(\kappa) = \frac{1}{\pi}\int_0^1 t^{-1/2}(1-t)^{-1/2}(1-\kappa t)^{\gamma+1+j}\mathrm{d}t \tag{S23}$$

(the second after $t \to \kappa t$), Euler's integral (Olver et al., 2010, 15.6.1) gives $H_j = B(\frac{1}{2}, \gamma+2+j)\,{}_2F_1(\frac{1}{2},\frac{1}{2};\gamma+\frac{5}{2}+j;k)$ and $G_j = {}_2F_1(-\gamma-1-j,\frac{1}{2};1;\kappa)$, and hence Equations (S4) and (S5). We write $\mathcal{F}_j = H_j$, $\Pi = -x^{\gamma+1}\sqrt{k}/(2(\gamma+1))$ across the limb and $\mathcal{F}_j = \pi G_j$, $\Pi = -x^{\gamma+1}/(2(\gamma+1))$ inside, so that $P_\gamma = \Pi\sum_j c_j\mathcal{F}_j$, and $P_{\gamma,J}$ for the sum through $j = J$.

S4.2 The remainder in j

Lemma 1 *Let a piece have $z_- \le z \le z_+$ with $z_- > p$ and $k \le \bar{k}$ (limb). With $\bar{x} = 1-(z_- - p)^2$ and $\bar{Q} = z_+^2 - p^2$,*

$$|P_\gamma - P_{\gamma,J}| \le E_\gamma^{(j)} = \frac{\bar{x}^{\gamma+1}\bar{Q}\,\bar{x}^{J+1}}{2(\gamma+1)(1-\bar{x})} \times \begin{cases} \sqrt{\bar{k}}\,H_{J+1}(\bar{k}) & \textit{across the limb,} \\ \pi & \textit{inside the disk.} \end{cases} \tag{S24}$$

Proof. For $j > J$, $|c_j| = |z^2-p^2|\,x^j$. The integrands of Equation (S23) are positive, and they do not increase with j because $0 \le 1-t \le 1$ and $0 \le 1-\kappa t \le 1$ for $0 \le \kappa \le 1$. Hence $0 < H_j \le H_{J+1}$ and $0 < \pi G_j \le \pi G_{J+1} \le \pi$ for $j > J$, and $|P_\gamma - P_{\gamma,J}| \le |\Pi|\,|z^2-p^2|\,\mathcal{F}_{J+1}\sum_{j>J}x^j = |\Pi|\,|z^2-p^2|\,\mathcal{F}_{J+1}x^{J+1}/(1-x)$. On the piece, x decreases and z^2-p^2 increases with $z > p$, k decreases with z across the limb, and H_{J+1} and $\sqrt{k}$ increase with k. Each factor is therefore bounded by its value at the appropriate end of the piece. For $\bar{k} = 1$, $H_{J+1}(1)$ follows from Gauss's sum (Olver et al., 2010, 15.4.20); for $\bar{k} < 1$ it equals $\hat{H}_{J+1}(\bar{k})$, which Lemma 4 bounds. □

S4.3 Analytic continuation of the geometry

Lemma 2 *(a) Across the limb, $z = p(1-2k) + W^{1/2}$ with $W = 1-4p^2k(1-k)$. On a closed disk D on which $4p^2|k|\,|1-k| < 1$, the principal root $W^{1/2}$ is analytic and agrees with Equation (S6) on $D\cap\mathbb{R}$, z has no zero, and there are branches of $(4pz)^{\gamma+1}$, of $x^{\gamma+1} = (4pzk)^{\gamma+1}$ and of $\sqrt{k}$ (the latter two if $0 \notin D$) that are analytic on D, agree with the real ones on $D\cap\mathbb{R}$, and satisfy $|f^{\gamma+1}| = |f|^{\gamma+1}$. (b) Inside the disk, $z = \big(V^{1/2} - p(2-\kappa)\big)/\kappa$ with $V = (\kappa-2p^2)^2 + h^2$, $h^2 = 4p^2(1-p^2)$. If $D = \{|\kappa-\kappa_0| \le r\}$, κ_0 real, does not contain 0 and*

$$\min_{-1\le v\le 1}\left[c^2 + h^2 - r^2 + 2crv + 2r^2v^2\right] > 0, \qquad c = \kappa_0 - 2p^2, \tag{S25}$$

the same conclusions hold with $x = 4pz/\kappa$.

Proof. (a) $|W-1| < 1$ implies $\mathrm{Re}\,W > 0$, so the principal root is analytic on D; it is positive on $D\cap\mathbb{R}$, as in Equation (S6). Since $\big(W^{1/2}+p(1-2k)\big)\big(W^{1/2}-p(1-2k)\big) = W - p^2(1-2k)^2 = 1-p^2 \ne 0$, z has no zero on D. A function without zeros on a disk has an analytic logarithm, unique up to $2\pi i$, which we fix by its real value on $D\cap\mathbb{R}$; then $|f^{\gamma+1}| = |\exp((\gamma+1)\log f)| = |f|^{\gamma+1}$. (b) Solving $\kappa = 4zp/(1-(z-p)^2)$ for z gives the stated root. $\mathrm{Re}\,V$ is harmonic, so its minimum over D lies on the circle $\kappa = \kappa_0 + re^{i\theta}$, where $\mathrm{Re}\,V$ is the bracket of Equation (S25) with $v = \cos\theta$. Thus $\mathrm{Re}\,V > 0$ on D, and the principal root is analytic and positive on $D\cap\mathbb{R}$. From

$\big(V^{1/2}-p(2-\kappa)\big)\big(V^{1/2}+p(2-\kappa)\big)=(1-p^2)\kappa^2$, a zero of $V^{1/2}-p(2-\kappa)$ requires $V=p^2(2-\kappa)^2$ and hence $\kappa=0\notin D$. The rest follows as in (a). □

On D the crude bounds $|z|\le p(1+2\max_D|k|)+(1+4p^2\max_D|k|\max_D|1-k|)^{1/2}$ (limb) and $|z|\le\big(d+r+p(2+\max_D|\kappa|)\big)/\min_D|\kappa|$ (inside, with $d=|\kappa_0-2p^2-ih|$, since $|V|\le(d+r)^2$) hold, together with $|x|\le 4p\max_D|z|\max_D|k|$ and $|x|\le 4p\max_D|z|/\min_D|\kappa|$.

Lemma 3 *Let f be analytic on $|u|\le\rho_2$ with $|f|\le C$ there, and $\rho<\rho_2$. Then*

$$\max_{|u|=\rho}|f(u)|\le\max_{0\le\ell<K}\big|f(\rho e^{2\pi i\ell/K})\big|+\frac{\pi\rho\,C}{K(\rho_2-\rho)}. \tag{S26}$$

Proof. For $|u|=\rho$, Cauchy's estimate on the disk of radius $\rho_2-\rho$ about u gives $|f'(u)|\le C/(\rho_2-\rho)$. Every point of the circle lies within arc length $\pi\rho/K$ of a sample, and $|f|$ changes along the arc by at most the arc length times $\max|f'|$. □

We apply Lemma 3 with $K=1024$ to z and x on a disk of radius $\rho_2>\rho$ that satisfies Lemma 2, with the crude bounds as C, and call the results Z and X. With $Q=p^2+Z^2$, on $|u|=\rho$

$$|c_0|\le C_0=1+Q,\qquad |c_j|\le C_j=Q\,X^j\quad(j\ge1). \tag{S27}$$

S4.4 Majorants of the Gauss functions

Lemma 4 *(a) Let $t_{n+1}/t_n=(n+a)(n+b)/((n+c)(n+1))$ and let $n+a$, $n+b$, $n+c>0$. Then for all $n'\ge n$*

$$\left|\frac{t_{n'+1}}{t_{n'}}\right|\le\bar r(n)=1+\frac{\max(0,a+b-c-1)}{n+c}+\frac{|ab-c|}{(n+c)(n+1)}, \tag{S28}$$

and if $q=\bar r(n)\,r<1$, then $\sum_{n'\ge n}|t_{n'}|r^{n'}\le|t_n|r^n/(1-q)$. (b) For $x\ge\frac16$, $\ln x-1/x\le\psi(x)\le\ln x$. (c) If $|1-\kappa|\le1$ and $\kappa\notin[1,\infty)$, then $|G_j(\kappa)|\le1$. (d) For $0<w<1$, $\max_{0<y\le w}y^m|\ln y|=L^*(w)$, with $L^*(w)=w^m|\ln w|$ for $w\le e^{-1/m}$ and $1/(em)$ otherwise.

Proof. (a) $(n+a)(n+b)-(n+c)(n+1)=n(a+b-c-1)+ab-c$ and $n\le n+1$ give the bound for $n'=n$; $\bar r$ does not increase with n. The tail is then dominated by a geometric series. (b) Binet's formula (Olver et al., 2010, 5.9.13) gives $\psi(x)=\ln x-1/(2x)-2\int_0^\infty t\,\mathrm{d}t/\big((t^2+x^2)(e^{2\pi t}-1)\big)$. The integral is positive and at most $x^{-2}\int_0^\infty t\,\mathrm{d}t/(e^{2\pi t}-1)=1/(24x^2)$, so $\ln x-1/(2x)-1/(12x^2)\le\psi(x)\le\ln x$, and $1/(12x^2)\le1/(2x)$ for $x\ge\frac16$. (c) On $t\in[0,1]$, $1-\kappa t$ lies on the segment from 1 to $1-\kappa$, so $|1-\kappa t|\le1$ and $1-\kappa t\ne0$; the weight in Equation (S23) integrates to 1. (d) $y^m|\ln y|$ increases on $(0,e^{-1/m}]$ and decreases on $[e^{-1/m},1)$. □

The series about $k=0$ have $a=b=\frac12$, $c=\gamma+\frac52+j$, and all their coefficients are positive, so $|H_j(k)|\le\hat H_j(|k|)$. About argument 1 the connection formulae (Olver et al., 2010, 15.8.4, 15.8.10) write $\mathcal{F}_j=A_j(u)+u^{m+j}\big[B_{l,j}(u)\ln u+B_{c,j}(u)\big]$ with $u=y$ or y'. For non-integer m, A_j and $u^{-m-j}(\mathcal{F}_j-A_j)$ are again hypergeometric series, with parameters $(a,b;1-m-j)$ and $(c-a,c-b;1+m+j)$, and $B_{l,j}=0$. For integer m (half-integer γ), A_j is a polynomial, $B_{l,j}$ is hypergeometric with parameters $(a+m+j,b+m+j;1+m+j)$, and $B_{c,j}$ has the coefficients of $B_{l,j}$ multiplied by $\psi_n=\psi(a+m+j+n)+\psi(b+m+j+n)-\psi(1+n)-\psi(1+m+j+n)$. All arguments of ψ here are at least $\frac12$. By Lemma 4(b) and $|\ln(X/Y)|\le|X-Y|/\min(X,Y)$, $|\psi_n|\le\Phi(n)$, where $\Phi(n)$ is built from terms $\alpha/(n+\beta)$ with $\alpha\ge0$, $\beta>0$ by sums and a maximum, and so does not increase with n. The tail of $\hat B_{c,j}$ after the nth term is therefore at most $\Phi(n+1)$ times that of $\hat B_{l,j}$. With Lemma 4(a), every $\hat F(r)$ and $\hat F^{>n_0}(r)$ is bounded by finitely many terms and a geometric tail.

S4.5 Truncation in n

Lemma 5 *Let $f = \tilde{f} + e$ with f analytic on $|u| \le \rho$ and $|f| \le M$ on $|u| = \rho$, let $\tilde{f}_n$ and e_n be the Taylor coefficients of $\tilde{f}$ and e, and let $\sum_{n \le N_B} |e_n| w^n \le \eta'$. Then for $|u| \le w < \rho$ and $N \le N_B$,*

$$\left| f(u) - \sum_{n \le N} \tilde{f}_n u^n \right| \le \sum_{n=N+1}^{N_B} |\tilde{f}_n|\, w^n + M\, \frac{(w/\rho)^{N_B+1}}{1 - w/\rho} + \eta'. \tag{S29}$$

Proof. $f(u) - \sum_{n \le N} \tilde{f}_n u^n = \sum_{n \le N_B} e_n u^n + \sum_{N < n \le N_B} \tilde{f}_n u^n + \sum_{n > N_B} f_n u^n$, and Cauchy's estimate gives $|f_n| \le M\rho^{-n}$. □

For the first-contact and second-contact pieces the computed coefficients are those of f itself ($e = 0$): they are obtained by multiplying and composing power series about $u = 0$, which is exact up to order $N_B = 64$. For the middle pieces, the series of $\mathcal{F}_j$ about $k = 0$ or $y = 0$ are truncated after n_t terms and re-centred exactly, so that $\tilde{f}$ is f with $\mathcal{F}_j$ replaced by its truncation and $e = \Pi \sum_j c_j(\mathcal{F}_j - \tilde{\mathcal{F}}_j)$. If $|e| \le \eta$ on $|u| = \rho$, Cauchy's estimate gives $\eta' = \eta/(1 - w/\rho)$.

S4.6 Proof of Theorem 1

Let the closed disk $|u| \le \rho_2$ satisfy Lemma 2, and let $w < \rho < \rho_2$, with $\rho < 1$ ($\rho < k_0$ and $k_0 + \rho < 1$ for the middle pieces in k; $\rho \le \kappa_0$ and $\kappa_0 + \rho < 1$ for those in κ). Then $P_{\gamma,J}$ and its parts are analytic on $|u| \le \rho$: Π and c_j by Lemma 2, and the Gauss series because their arguments stay in the unit disk. On $|u| = \rho$ we have $|x^{\gamma+1}| \le X^{\gamma+1}$, $|(4pz)^{\gamma+1}| \le (4pZ)^{\gamma+1}$ and $|\sqrt{k}| \le \sqrt{\max|k|}$, and C_j from Equation (S27). With $\Pi^* = X^{\gamma+1}/(2(\gamma+1))$, Equation (S29) applies to the following parts.

First contact. $P_{\gamma,J} = k^m R(k)$ with $R = -(4pz)^{\gamma+1}/(2(\gamma+1)) \sum_{j \le J} c_j H_j$ and $M = (4pZ)^{\gamma+1}/(2(\gamma+1)) \sum_j C_j \hat{H}_j(\rho)$. The error is w^m times the right-hand side of Equation (S29) with $\eta' = 0$.

Middle pieces in k. $f = P_{\gamma,J}$, $M = \Pi^* \sqrt{k_0 + \rho} \sum_j C_j \hat{H}_j(k_0 + \rho)$ and $\eta = \Pi^* \sqrt{k_0 + \rho} \sum_j C_j \hat{H}_j^{>n_t}(k_0 + \rho)$.

Middle pieces in κ. $f = P_{\gamma,J}$ and $M = \pi\Pi^* \sum_j C_j$ by Lemma 4(c), since $|1 - \kappa| \le 1 - \kappa_0 + \rho \le 1$. With $r_y = 1 - \kappa_0 + \rho < 1$, $\operatorname{Re} y \ge 1 - \kappa_0 - \rho > 0$ and $|\ln y| \le \Lambda = \max\big(|\ln(1 - \kappa_0 - \rho)|, |\ln r_y|\big) + \pi/2$,

$$\eta = \Pi^* \sum_j C_j \left[\hat{A}_j^{>n_t}(r_y) + r_y^{m+j} \big(\Lambda\, \hat{B}_{l,j}^{>n_t}(r_y) + \hat{B}_{c,j}^{>n_t}(r_y) \big) \right]. \tag{S30}$$

Second contact. $P_{\gamma,J} = R + u^m(S_L \ln u + S_C)$ with $R = \Pi \sum_j c_j A_j$, $S_L = \Pi \sum_j c_j u^j B_{l,j}$ and $S_C = \Pi \sum_j c_j u^j B_{c,j}$, and $M_R = \Pi^\dagger \sum_j C_j \hat{A}_j(\rho)$, $M_L = \Pi^\dagger \sum_j C_j \rho^j \hat{B}_{l,j}(\rho)$, $M_C = \Pi^\dagger \sum_j C_j \rho^j \hat{B}_{c,j}(\rho)$, where $\Pi^\dagger = \Pi^* \sqrt{1 + \rho}$ outside and Π^* inside. By Lemma 4(d) the error is at most the bound of Equation (S29) for R, plus $L^*(w)$ times that for S_L, plus w^m times that for S_C.

In each case $E_\gamma^{(n)}$ is the resulting bound and $P_{\gamma,J} - \hat{P}_\gamma$ is at most $E_\gamma^{(n)}$; Lemma 1 adds $E_\gamma^{(j)}$. The implementation stores $\sum_i c_i \hat{P}_{\gamma_i}$. For parts shared by all exponents (R and the middle pieces) it uses $\rho = \min_i \rho_i$ and $M = \sum_i |c_i| M_i$, which is valid by the maximum principle; the parts carrying u^{m_i} are bounded separately for each exponent. Dividing $\sum_i c_i P_{\gamma_i}$ by the stellar flux $2\pi \sum_i c_i/(\alpha_i + 2)$ gives the flux, which completes the proof. □

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